INNOPOLIS UNIVERSITY

Bachelor of Science in Computer Science

ViKi: Vision-Based Kinematic Imitation for Object-Centric Retargeting of Human Manipulation Demonstrations to Robot Manipulators

Bachelor’s Graduation Thesis

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Abstract

Imitation learning for robot manipulation is bottlenecked by the cost of demonstration data: teleoperation requires dedicated hardware and an operator per episode, which limits both throughput and the diversity of recorded behaviour. Passive capture of bare-handed human demonstrations removes that bottleneck, but existing passive pipelines either discard the hand entirely and plan over sparse object keyframes, or reconstruct a full parametric hand mesh from monocular video and retarget it in two decoupled stages—smoothing the observed trajectory, solving inverse kinematics, then smoothing the joint output again. The latter pattern is unsound: pre-filtering in task space cannot guarantee smoothness of the nonlinear inverse-kinematics solution, and it attenuates genuine fast motion together with perception noise. Optimisation-based retargeting that fuses data fidelity, temporal smoothness and joint limits into a single cost functional is by now standard—but only in *online teleoperation* systems, where a human remains in the loop and no dataset is produced. This thesis closes that gap. It presents ViKi, an offline pipeline that captures bare-handed manipulation demonstrations with a metrically calibrated multi-camera RGB-D rig anchored to a ChArUco workspace frame, represents each demonstration as a dense object-relative wrist trajectory in $\mathrm{SE}(3)$ with a binary gripper state, and retargets it onto a robot manipulator by minimising a single weighted cost functional that combines a perception-confidence-weighted data term, a velocity-smoothness penalty and an acceleration penalty subject to configuration, velocity and collision constraints. Synthesised trajectories are replayed on the physical manipulator before the dataset is finalised, so that kinematically infeasible episodes are removed at the source rather than surfacing as policy failures later. The system was evaluated on a constrained class of tabletop pick-and-place and pouring tasks with a UR3 manipulator, two Azure Kinect DK cameras and one Intel RealSense D435i. The proposed formulation reduced wrist-pose tracking error by **XX%** and improved joint-space smoothness by **XX%** relative to a sequential smooth–IK–smooth baseline, while replay screening rejected **XX%** of episodes as infeasible; policies trained on the resulting LeRobot-compatible dataset reached **XX%** success. The results indicate that the online-teleoperation retargeting formulation transfers to the offline dataset-generation regime, and that closing the embodiment gap by physical replay before export is a practical alternative to filtering after training.

Keywords: imitation learning; motion retargeting; inverse kinematics; RGB-D calibration; object-centric representation; robot learning from human video.

Аннотация

Обучение роботов манипуляции методом имитации ограничено стоимостью получения демонстраций: телеоперация требует специализированного оборудования и оператора на каждый эпизод, что ограничивает и производительность записи, и разнообразие поведения. Пассивный захват демонстраций, выполняемых голой рукой человека, снимает это ограничение, однако существующие пассивные пайплайны либо полностью отбрасывают кисть и планируют по разреженным ключевым кадрам объекта, либо восстанавливают полную параметрическую mesh-модель кисти из монокулярного видео и выполняют ретаргетинг в две развязанные стадии: сглаживание наблюдаемой траектории, решение обратной задачи кинематики, повторное сглаживание в пространстве суставов. Последняя схема методологически некорректна: предварительная фильтрация в пространстве задачи не гарантирует гладкости нелинейного решения обратной кинематики и подавляет истинное быстрое движение вместе с шумом восприятия. Постановка ретаргетинга как единого функционала стоимости, объединяющего точность соответствия данным, временную гладкость и ограничения на суставы, к настоящему моменту стала стандартом — но только в системах *онлайн-телеоперации*, где человек остаётся в контуре управления и датасет не формируется. Настоящая работа закрывает этот разрыв. Предлагается ViKi — офлайн-пайплайн, который захватывает демонстрации манипуляции голой рукой при помощи метрически откалиброванного мультикамерного RGB-D рига, привязанного к системе координат рабочей области по маркерной доске ChArUco, представляет каждую демонстрацию как плотную объектно-относительную траекторию запястья в $\mathrm{SE}(3)$ с бинарным состоянием захвата и выполняет ретаргетинг на роботизированный манипулятор путём минимизации единого взвешенного функционала, объединяющего член соответствия данным со взвешиванием по достоверности восприятия, штраф на скорость и штраф на ускорение при ограничениях на конфигурацию, скорость и столкновения. Синтезированные траектории воспроизводятся на физическом манипуляторе до финализации датасета, благодаря чему кинематически нереализуемые эпизоды отсеиваются у источника, а не проявляются позднее как отказы политики. Система оценена на ограниченном классе настольных задач захвата–переноса и переливания с манипулятором UR3, двумя камерами Azure Kinect DK и одной Intel RealSense D435i. Предложенная постановка снизила ошибку отслеживания позы запястья на **XX%** и повысила гладкость в пространстве суставов на **XX%** относительно последовательной базовой схемы smooth–IK–smooth, при этом отбор по результатам воспроизведения отклонил **XX%** эпизодов как нереализуемые; политики, обученные на полученном LeRobot-совместимом датасете, достигли успешности **XX%**. Результаты показывают, что формулировка ретаргетинга, разработанная для онлайн-телеоперации, переносима в режим офлайн-генерации датасета, а закрытие разрыва эмбодиментов через физическое воспроизведение до экспорта является практичной альтернативой фильтрации после обучения.

Ключевые слова: обучение с имитацией; ретаргетинг движения; обратная кинематика; калибровка RGB-D; объектно-центричное представление; обучение роботов по видео человека.

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[Placeholder.] The author thanks **[supervisor]** for supervision and guidance throughout this work, and the Innopolis Robotics Society for maintaining a parallel development fork of the system, whose independent implementation attempts informed several of the architectural decisions reported here.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

1.1 Background

Visuomotor policies such as Action Chunking Transformers (ACT) and Diffusion Policy have made behaviour cloning a practical route to robot manipulation, but their performance scales with the quantity and quality of demonstration data. The dominant way of acquiring that data is teleoperation: a human operator drives the robot through the task while joint states and camera streams are recorded. Teleoperation yields action labels that are, by construction, exactly executable on the target embodiment, which is its principal advantage. Its disadvantages are equally structural. Each episode consumes an operator's time in real time, the required hardware—a VR controller, an exoskeleton or a haptic device—is expensive and embodiment-specific, and the resulting motion is shaped as much by the latency and ergonomics of the interface as by the task itself.

Human demonstrations recorded passively, with the demonstrator using their own hands and no interface at all, invert this trade-off. Data collection becomes cheap, fast and natural; a demonstrator can record dozens of episodes in the time a teleoperator records one, and the recorded motion reflects how the task is actually performed rather than how it can be performed through a control interface. What is lost is precisely the property teleoperation guarantees: there are no action labels, and the kinematics of a human arm and hand do not correspond to the kinematics of a robot manipulator. Bridging that gap is the retargeting problem, and it is the subject of this thesis.

1.2 Problem Statement

Passive capture makes demonstration data cheap but produces observations, not actions. Converting an observed human hand trajectory into a robot joint trajectory that (i) reproduces the task-relevant motion, (ii) is smooth enough for a policy to imitate, and (iii) is physically executable on the target manipulator, is an underdetermined problem contaminated by perception noise.

Two failure modes dominate. The first is noise amplification through the kinematic chain. Monocular or neural depth estimates of hand keypoints are noisy at the centimetre scale; inverse kinematics is nonlinear, so a bounded task-space perturbation may map to an unbounded joint-space excursion near singularities. The widespread response—low-pass filtering the observed trajectory before solving inverse kinematics, then filtering the joint trajectory afterwards—does not solve this. Task-space pre-filtering is applied in the wrong space to guarantee joint-space smoothness, and it removes genuine high-frequency task content along with the noise. This is the garbage-in-garbage-out pattern that motivates the present work.

The second failure mode is representational. A trajectory recorded as an absolute sequence of positions in a camera or world frame is tied to the exact spatial layout of the recording session. If the object is moved, the trajectory is wrong; if the embodiment changes, it is wrong in a different way. What generalises is the relation between the hand and the object being manipulated, not the absolute path of the hand through space.

1.3 Research Gap

Both failure modes have known remedies, and the literature has adopted them— but in disjoint communities.

Optimisation-based retargeting, in which data fidelity, temporal smoothness and joint limits are posed as a single constrained optimisation over the robot configuration, is standard in *online teleoperation*. AnyTeleop and the associated `dex-retargeting` library established the canonical form; Open-TeleVision, Bunny-VisionPro and ACE build directly on it; DexFlow extends the smoothness penalty to second order. In every case, however, a human remains in the control loop, the optimisation runs at interactive rates, and no dataset is produced.

Conversely, *offline* pipelines that generate datasets from passive human video adopt object-centric representations but not the joint-optimisation formulation. ORION plans over sparse object keyframes and does not perform per-frame hand retargeting at all. OKAMI reconstructs a parametric body-and-hand model and retargets in two decoupled stages, warping a reference trajectory to a new object position and then solving inverse kinematics—structurally the same sequential pattern criticised in Section 1.2.

A targeted prior-art review conducted for this work (Chapter 2, Section 2.7) found no published system combining passive markerless capture, a metrically calibrated multi-camera RGB-D rig, an object-centric dense trajectory representation, single-cost-functional retargeting, imitation-learning-ready dataset export, and physical replay used as a pre-export validation step. The nearest systems each satisfy some criteria and miss others. The gap addressed here is therefore not the invention of a new retargeting operator, but the transfer of an established formulation into a regime where it has not been applied, together with the sensing and validation infrastructure that regime requires.

1.4 Research Questions and Hypotheses

The work is guided by one primary research question and three sub-questions.

RQ. *How can human manipulation demonstrations captured passively by a calibrated multi-camera RGB-D rig be retargeted onto a robot manipulator so as to yield datasets suitable for training visuomotor imitation policies?*

RQ1. How does formulating retargeting as a single weighted cost functional, combining a confidence-weighted data term with velocity and acceleration penalties, affect the fidelity and smoothness of the resulting joint trajectories compared with a sequential smooth–IK–smooth pipeline?

RQ2. What accuracy can be achieved for wrist pose estimation by confidence-weighted fusion of multiple calibrated RGB-D views anchored to a common ChArUco workspace frame, and how does the residual error decompose across its independent sources?

RQ3. To what extent does replaying synthesised trajectories on the physical manipulator, prior to dataset export, improve the executability of the exported dataset?

The corresponding hypotheses are stated below. Following the convention for experimental computer-science hypotheses, each identifies an independent variable, a dependent variable, a population and a relation.

H1 (behaviour).
Retargeting posed as a single weighted cost functional produces joint trajectories with lower task-space tracking error and higher smoothness than a sequential smooth–IK–smooth pipeline, on the same recorded demonstrations. *Independent variable:* retargeting formulation (joint vs. sequential). *Dependent variables:* wrist-pose tracking RMSE; spectral arc length of the joint trajectory. *Population:* recorded demonstrations of the constrained task class defined in Section 4.2.2. *Relation:* the joint formulation dominates on both measures.

H2 (behaviour).
Weighting the data term by per-frame perception confidence reduces tracking error relative to uniform weighting, and the reduction is larger on frames where hand landmarks are partially occluded. *Independent variable:* presence of confidence weighting. *Dependent variable:* wrist-pose tracking RMSE, stratified by occlusion. *Population:* as above. *Relation:* monotone reduction, amplified under occlusion.

H3 (dependability).
Screening trajectories by physical replay before export increases the proportion of exported episodes that execute successfully on the manipulator. *Independent variable:* presence of the replay screening stage. *Dependent variable:* executable fraction of the exported dataset. *Population:* as above. *Relation:* positive.

The questions were checked against the FINER criteria. They are *feasible*, in that the hardware, solver and evaluation protocol are all available and the task class is deliberately constrained; *interesting* and *relevant* to the robot-learning community, because demonstration cost is a recognised bottleneck; *novel* in the specific sense established in Section 1.3, namely a regime transfer rather than a new operator; and *ethical*, as the study involves no human subjects beyond the author performing demonstrations and no personally identifying data is retained.

1.5 Contributions

This thesis makes the following contributions.

1. A retargeting formulation for the *offline* setting that poses data fidelity, velocity smoothness and acceleration smoothness as one constrained optimisation over the robot configuration, with the data term weighted by per-frame perception confidence and made robust by a Huber loss (Section 3.7).

2. An object-centric dense trajectory representation that stores each demonstration both relative to the manipulated object and relative to a fixed ChArUco workspace anchor, permitting downstream consumers to select the representation that generalises better (Section 3.6).

3. A multi-camera RGB-D capture and calibration procedure with confidence-weighted depth fusion and an explicit treatment of cross-device timestamp alignment, together with an error budget that separates the three independent contributions to residual pose error (Sections 3.3–3.5, Section 5.3).

4. A physical-replay validation stage that filters kinematically infeasible trajectories before dataset export, and an empirical characterisation of what that stage rejects (Sections 3.8, 4.7).

5. An open implementation exporting LeRobot-compatible datasets, evaluated end-to-end by training ACT and Diffusion Policy on the generated data (Chapter 4).

1.6 Thesis Structure

Chapter 2 reviews prior work on hand representation, retargeting formulations, task representation, multi-camera RGB-D calibration and dataset generation, and positions ViKi against the nearest published systems. Chapter 3 describes the design of the system and the derivation of the retargeting cost functional. Chapter 4 reports the implementation, the evaluation protocol and the experimental results. Chapter 5 interprets those results, decomposes the error budget and states the limitations. Chapter 6 concludes and outlines future work.

Chapter 2

Literature Review

2.1 Preamble

This chapter surveys the state of knowledge relevant to converting passively recorded human manipulation demonstrations into robot-executable datasets. It is organised not chronologically but by the architectural elements of the proposed system, so that for every design decision taken in Chapter 3 the alternatives considered and the reason for the choice are visible.

Section 2.2 examines how the human hand is represented, from full parametric meshes to the minimal wrist-pose-plus-gripper abstraction. Section 2.3 contrasts sequential and single-cost-functional retargeting formulations and reviews the solvers used to realise them. Section 2.4 compares object-centric and world-frame task representations. Section 2.5 covers multi-camera RGB-D calibration, depth fusion and cross-device synchronisation, with emphasis on reported quantitative accuracy. Section 2.6 reviews dataset formats, downstream policies and demonstration curation. Section 2.7 positions ViKi against the nearest published systems, and Section 2.8 states the resulting methodological choices.

Sources were drawn from arXiv (`cs.RO`, `cs.CV`, `cs.LG`), the proceedings of CoRL, RSS, ICRA, IROS, CVPR and NeurIPS, peer-reviewed sensor and calibration literature, and vendor documentation for the depth cameras and solvers employed. Where a source is a recent preprint whose peer-review status could not be confirmed, this is stated explicitly.

2.2 Representing the Human Hand

2.2.1 Full parametric reconstruction

A large body of work recovers the complete kinematics of the demonstrator. OKAMI [10] combines SMPL-H for the body with MANO for the hand and retargets the two separately. DexCanvas [23] builds a human–robot correspondence dataset by fitting MANO to data from twenty-two infrared motion capture cameras and two RGB-D sensors. DexMV [11] likewise recovers a full MANO hand together with object pose from video before translating the demonstration into simulation.

Full reconstruction yields a rich representation—the pose of every finger joint—and is necessary when the target embodiment is a multi-fingered dexterous hand. It also introduces a heavy perception stack and an additional, poorly characterised error source: the mesh fit itself. For a parallel-jaw gripper, most of that representational richness is discarded downstream.

2.2.2 Minimal representation: wrist pose and gripper state

The alternative, adopted in this work, represents the hand as a wrist pose in $\mathrm{SE}(3)$ together with a scalar or binary gripper state. This is not a concession made for tractability; it is the representation used by the mainstream of current teleoperation systems. Open-TeleVision [2] maps the operator's wrist pose to the arm end-effector through closed-loop inverse kinematics and encodes a parallel gripper by the single thumb–index vector. Bunny-VisionPro [3] and ACE [4] adopt the same decomposition, ACE additionally demonstrating that it transfers across dexterous hands, parallel grippers and quadrupeds.

Methodological support for down-weighting the wrist comes from the ablation study of Xin et al. [7], who assemble a comprehensive retargeting objective—global hand pose, overall hand shape, relative fingertip positions and fingertip orientations—and remove terms one at a time. They find that precise tracking of the global wrist pose is frequently unnecessary and can degrade fingertip accuracy, and consequently replace the wrist-position objective with a thumb-tip objective while regularising wrist orientation only weakly. The implication for a wrist-plus-gripper representation is that the wrist target should enter the optimisation as a weighted soft task rather than a hard constraint.

2.2.3 Egocentric and monocular capture

Several systems reduce capture-hardware requirements further. EgoMimic [16] records bare-handed demonstrations through Project Aria glasses and co-trains on human and robot data. DexWild [17] collects in-the-wild bare-hand data through a low-cost portable device and likewise co-trains. A recent egocentric RGB-D system [22] generates trajectories offline using analytic damped-least-squares inverse kinematics and replays them on a low-cost arm. These works confirm a trend toward lighter capture rigs, but each accepts monocular or single-view geometry, and none combines it with the metric multi-view accuracy that the present work requires.

2.3 Retargeting Formulations

2.3.1 The single cost functional

The prevailing modern formulation poses retargeting as a constrained nonlinear least-squares or quadratic program over the robot configuration $\boldsymbol{q}_t$ at each timestep, minimising a weighted sum of a data-fidelity term and a temporal smoothness term subject to joint limits. AnyTeleop [1] gives the canonical instance: the objective sums, over keypoints, the squared residual between a scaled human keypoint vector and the corresponding robot forward-kinematics vector, plus a first-order penalty on configuration change between consecutive frames. The accompanying `dex-retargeting` library exposes a vector optimiser for real-time teleoperation, a position optimiser explicitly recommended for offline dataset processing, and a contact-aware DexPilot optimiser; it is built on Pinocchio and solved with sequential quadratic programming.

Two refinements are directly relevant here. First, per-term confidence weighting: in the formulation reported by Open-TeleVision [2], each keypoint residual is scaled by a distance-dependent switching weight that increases when a pinch is detected. The mechanism is precisely analogous to weighting by a per-frame perception confidence, and it establishes that weighting individual data terms by a scalar signal is accepted practice. Second, second-order smoothness: DexFlow [5] augments the standard objective with a dynamic smoothing weight and Hessian regularisation to enforce C^2 continuity, penalising curvature and not merely velocity. Robustification by a Huber loss on the data term, reported in recent dexterous teleoperation work [6], provides a mechanism for rejecting perception outliers inside the optimiser rather than smoothing them away beforehand.

2.3.2 The sequential pattern and its critique

Against this, the sequential pattern—filter the observed task-space trajectory, solve inverse kinematics per frame, filter the joint output— remains common in offline video-to-robot pipelines. Its deficiency is structural rather than empirical. Filtering is applied in task space, whereas the smoothness requirement is on the joint trajectory; because inverse kinematics is nonlinear, a smooth Cartesian input does not imply a smooth joint output, particularly near singularities. Pre-filtering also attenuates genuine fast motion indiscriminately with noise. In the single-functional formulation the smoothness penalty acts directly on $\boldsymbol{q}_t$, so joint-space smoothness is obtained by construction.

2.3.3 Offline pipelines and their retargeting operators

ORION [9] does not perform per-frame hand retargeting: it constructs an object-centric plan over sparse keyframes from an open-world object graph. OKAMI [10] warps a reference trajectory to a new object position and then solves inverse kinematics—the sequential pattern. HOWTransfer [19], the closest published pipeline in workflow terms, performs passive markerless capture and transfers trajectories using flow-matching $\mathrm{SE}(3)$ grasp sampling with Laplacian trajectory editing, not a single weighted cost functional; its representation is hand-centric rather than object-centric. No offline pipeline located in this review adopts the AnyTeleop-style single functional.

2.3.4 Solvers

Pinocchio underlies the retargeting stack of AnyTeleop, Open-TeleVision and ACE. Pink [8] exposes weighted inverse kinematics directly as a quadratic program: each task contributes a quadratic residual with a weight matrix normalising task coordinates to common units, while joint limits, velocity limits and self-collision barriers enter as inequality constraints, optionally with control-barrier formulations. This structure admits the entire proposed cost functional natively—a frame task for the wrist target, posture and velocity regularisation tasks for smoothness, and limits as constraints— solved as one program per frame. The solver choice made in Chapter 3 therefore lies on the mainstream path rather than diverging from it.

2.4 Task Representation

Object-centric representations dominate recent work on cross-environment and cross-embodiment generalisation. ORION [9] represents a task as the three-dimensional object trajectory relative to initial and final object positions, arguing that this generalises across visual backgrounds, camera shifts, spatial layouts and novel object instances, and that full-body reconstruction of the demonstrator is unnecessary. SPOT [12] encodes each task as the $\mathrm{SE}(3)$ object-pose trajectory relative to the target explicitly in order to decouple embodiment-specific actions from sensory input, enabling learning from action-free human demonstrations, and conditions a diffusion policy on the object trajectory. Flow-based formulations [13] make an equivalent argument while relaxing the rigid-object assumption that pose-trajectory methods require.

OKAMI [10] occupies an intermediate position: the representation is demonstrator-centric, but object-aware warping adapts trajectories to new object positions—an implicit acknowledgement that pure world-frame replay fails when the object moves. Vi-RoIL [20] is object-centric by construction, expressing wrist trajectories and keyframe hand poses in the object frame, but recovers metric scale from a monocular view of a marker cube and evaluates only in simulation.

The trade-off is that object-centric representations require robust object pose or segmentation, and pose-trajectory variants assume rigidity, whereas world-frame trajectories are simpler but tied to the recording layout. Storing both, as proposed in Section 3.6, defers the choice to the downstream consumer at negligible storage cost.

2.5 Multi-Camera RGB-D Calibration, Fusion and Synchronisation

2.5.1 Fiducial choice

ChArUco boards combine the sub-pixel corner accuracy of a checkerboard with the robust identification and occlusion tolerance of ArUco markers. A 6×8 board provides up to thirty-five corners per view against four for a single AprilTag, and remains detectable under partial occlusion and at the image boundary where a plain checkerboard fails. Reported reprojection errors for ChArUco and ArUco calibration are typically below half a pixel. AprilTags are easier to detect at long range and oblique incidence and suffice for extrinsic-only estimation, but are ill-suited to intrinsic calibration owing to the small number of corners. Current practice is therefore to use ChArUco for intrinsics and either fiducial for extrinsics, preferring AprilTags for oblique overhead views.

2.5.2 Reported accuracy of multi-Azure-Kinect rigs

Romeo et al. [24] provide the cleanest direct comparison of calibration procedures for multi-Kinect rigs, reporting that three-dimensional procedures outperform two-dimensional chessboard approaches, with mean distances between corresponding point-cloud points of 21.4 mm at colour resolution and 9.9 mm at infrared resolution for a static scene, and 20.9 mm and 7.4 mm respectively for a dynamic scene; body-joint alignment error was 35.4 mm. Darwish et al. [25] calibrate five Azure Kinects against a ChArUco-coded cube, combining reprojection error with point-to-plane and patch-to-plane terms to compensate depth noise. Kim et al. [26] perform coarse registration from body tracking followed by feature-match refinement, retaining above ninety-five per cent inliers. Bu and Lee [27] use markerless coarse calibration with coloured ICP refinement and report—importantly for rig geometry—that ICP fails for near-opposing camera placements owing to insufficient point-cloud overlap.

Single-sensor specifications bound the achievable accuracy from below. The Azure Kinect DK is specified at a random error standard deviation of at most 17 mm and a typical systematic error below 11 mm plus 0.1 per cent of distance [30]; independent evaluation confirms these figures [28], and comparative work finds improved spatial and temporal accuracy relative to Kinect v2 beyond 2.5 m [29]. The Intel RealSense D435i is specified at depth accuracy below one per cent of distance [31]. The practical consequence is that centimetre-level, not millimetre-level, agreement should be expected between calibrated views.

2.5.3 Depth fusion

Naive averaging of overlapping depth observations is inferior to confidence-weighted fusion. Weighting by distance and by the angle between the surface normal and the viewing direction—so that surfaces observed near perpendicular contribute more than obliquely observed ones—is established practice [34]. Learned confidence maps estimated from the depth signal itself are used to re-weight depth features before fusion [35], and surfel-based reconstruction attaches an explicit confidence value to each element and performs curvature-weighted registration [36]. Both sensor families used in this work expose per-pixel invalidation or confidence signals that can drive such weights directly.

2.5.4 Cross-device synchronisation

Azure Kinect devices support hardware synchronisation through daisy-chained or star-wired trigger cabling, with subordinate devices required to start before the master and depth exposures staggered to prevent laser interference; the vendor documentation states explicitly that equal timestamps between depth and colour streams, or between cameras, are not guaranteed [30]. RealSense devices provide an analogous inter-camera synchronisation mode, with the counter-intuitive property that timestamps appear aligned when hardware synchronisation is disabled and exhibit bounded drift when it is enabled, the drift itself serving as verification [31, 32].

The pitfall for a heterogeneous rig is that hardware clocks on different device families are not mutually comparable, since each counts its own cycles from its own epoch. Documented practice for such rigs is to stamp every arriving frame with a single host-side monotonic clock and to align streams by nearest timestamp in that common domain, accepting millisecond-scale precision [32]. Because Azure Kinect and RealSense triggers are mutually incompatible, this is the only available mechanism for aligning the two families, and it is the mechanism adopted in Section 3.4.

2.5.5 Hand–eye calibration

The OpenCV `calibrateHandEye` routine implements the Tsai–Lenz, Park–Martin, Horaud–Dornaika, Andreff and Daniilidis solvers. A comparative noise study [33] finds the Daniilidis, Chou and Park methods most robust to translation noise, with Daniilidis yielding the best translation estimates, while Tsai, Li and Lu degrade as rotation noise grows. Methods that decouple rotation from translation are faster but propagate rotation error into the translation estimate; Daniilidis solves both simultaneously through dual quaternions. Because the present system anchors cameras and robot to a shared ChArUco workspace frame, classical hand–eye estimation is largely bypassed; where an explicit camera-to-robot transform is required, Daniilidis is used.

2.6 Dataset Generation and Downstream Policies

The LeRobot HDF5 conventions together with ACT and Diffusion Policy constitute the default imitation-learning toolchain and define the export target of this work. OpenVLA-OFT [15] reports that parallel decoding, action chunking, a continuous action representation and $L1$ regression raise average success across four task suites from 76.5 to 97.1 per cent while increasing action-generation throughput twenty-six-fold, and is therefore a credible alternative consumer of the generated data.

Curation matters as much as generation. Demo-SCORE [14] trains a classifier on policy rollouts to filter low-quality demonstrations and reports an absolute success-rate improvement exceeding fifteen to thirty-five per cent over training on all demonstrations. Smoothness-based scoring of trajectories has been reported to yield comparable gains at a fraction of the data **[citation to be confirmed]**. This motivates treating dataset curation as an explicit pipeline stage rather than an afterthought, and—in the present design—moving it upstream of export in the form of physical replay screening.

2.7 Positioning of the Proposed System

Table I compares ViKi against the nearest published systems along the six criteria that jointly define the proposed architecture: (C1) passive markerless capture without teleoperation hardware; (C2) multi-camera metrically calibrated RGB-D sensing; (C3) object-centric dense trajectory representation; (C4) retargeting as a single confidence-weighted cost functional; (C5) export of an imitation-learning-ready dataset; and (C6) physical replay used to validate the dataset before export.

TABLE I
Comparison of the Proposed System Against the Nearest Published Work
Along the Six Criteria Defined in Section 2.7

System	C1	C2	C3	C4	C5	C6
ViKi (proposed)	✓	✓	✓	✓	✓	✓
HOWTransfer [19]	✓	∼	—	—	∼	✓
Vi-RoIL [20]	∼	—	✓	—	∼	—
ORION [9]	✓	∼	✓	—	—	—
OKAMI [10]	✓	∼	∼	—	∼	—
EgoMimic [16]	✓	—	—	—	∼	—
DexWild [17]	∼	—	—	—	∼	—
Tool-as-Interface [18]	✓	∼	∼	—	∼	—
RoboPaint [21]	—	✓	∼	—	✓	✓
DexCanvas [23]	—	✓	—	—	∼	—

✓ = satisfied; ∼ = partially satisfied; — = not satisfied.

No located system satisfies all six criteria simultaneously. HOWTransfer [19] is closest in workflow, satisfying C1 and C6, but is hand-centric rather than object-centric, uses calibrated stereo RGB rather than metric RGB-D, and does not employ a single cost functional. Vi-RoIL [20] is closest in representation, satisfying C3, but relies on a monocular view of a marker cube and is evaluated only in simulation. The intersection of the online joint-optimisation retargeting literature with the offline object-centric dataset-generation literature is therefore unoccupied, and it is that intersection which this work targets.

Several of the systems listed are recent preprints [19, 21, 22, 23] whose peer-review status could not be confirmed at the time of writing; their reported results are treated as provisional.

2.8 Conclusion of the Review

The review supports the following methodological choices, which Chapter 3 implements.

1. *Hand representation.* A wrist pose in SE(3) with a binary gripper state is adopted, following [2, 3, 4], and the wrist target enters as a weighted soft task following the ablation evidence of [7]. Parametric mesh reconstruction is rejected as unnecessary for a parallel-jaw target and as an additional error source.

2. *Retargeting.* A single weighted cost functional is adopted, following [1, 5], with confidence weighting after [2] and Huber robustification after [6]. The sequential smooth–IK–smooth pattern is rejected on the structural grounds set out in Section 2.3.

3. *Solver.* Pink on Pinocchio [8] is adopted, since it expresses the entire functional natively as a constrained quadratic program and is the solver family used by the reference teleoperation systems.

4. *Task representation.* Object-relative trajectories are adopted following [9, 12], and stored alongside workspace-anchored trajectories so that the choice is deferred downstream.

5. *Sensing.* ChArUco workspace anchoring is adopted; centimetre-level cross-camera agreement is assumed on the evidence of [24]; camera placement preserves substantial view overlap on the evidence of [27]; depth is fused with distance, incidence-angle and sensor confidence weights following [34, 35].

6. *Synchronisation.* The two Azure Kinects are hardware synchronised; the RealSense, which cannot share their trigger, is aligned through a single host-side monotonic clock following [32].

7. *Curation.* Dataset curation is treated as an explicit stage and moved upstream of export in the form of physical replay screening, motivated by [14] and by the replay-based validation of [19].

Chapter 3

Design and Methodology

3.1 System Overview

ViKi is an offline pipeline: demonstrations are recorded, processed and exported in distinct stages, with no requirement that any stage run at interactive rates. This is a deliberate design choice. Relaxing the real-time constraint that shapes teleoperation systems permits global optimisation over whole trajectories, iterative refinement, and a validation stage that executes on physical hardware—none of which is available to an online system.

The pipeline comprises six stages, shown in Fig. 3.1.

1. *Capture.* Synchronised RGB-D streams are recorded from a calibrated multi-camera rig while a demonstrator performs the task bare-handed.

2. *Perception.* Hand landmarks are detected per view, lifted to three dimensions using sensor depth, and fused across views with confidence weights to yield a wrist pose in $\mathrm{SE}(3)$ and a gripper state.

3. *Representation.* The wrist trajectory is expressed both relative to the manipulated object and relative to the ChArUco workspace anchor.

4. *Retargeting.* A robot joint trajectory is obtained by minimising a single weighted cost functional subject to kinematic and collision constraints.

5. *Replay validation.* The synthesised trajectory is executed on the physical manipulator; proprioception is recorded and infeasible episodes are rejected.

6. *Export.* Surviving episodes are written in a LeRobot-compatible format for downstream policy training.

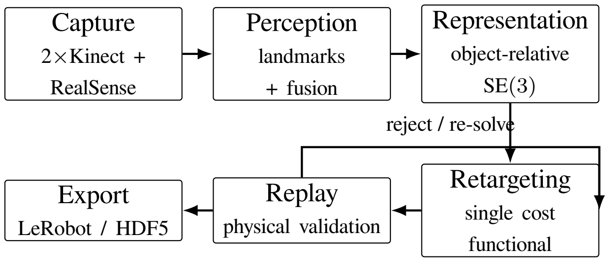

Fig. 3.1. Stages of the ViKi pipeline. Capture, perception and representation are embodiment-agnostic; retargeting, replay validation and export are specific to the target manipulator. The feedback edge indicates that episodes failing replay are either re-solved with modified weights or discarded.

3.2 Hardware and Workspace

The capture rig comprises two Azure Kinect DK cameras and one Intel RealSense D435i observing a tabletop workspace, with a UR3 manipulator mounted at a fixed position relative to that workspace. The Azure Kinect was selected for the primary views because its time-of-flight depth sensing provides substantially better range estimation than neural regression from RGB, and because its specified error characteristics—random error standard deviation at most 17 mm and systematic error below 11 mm plus 0.1 per cent of distance [30]—are documented and independently verified [28]. The RealSense provides a supplementary close-range view at higher depth accuracy [31].

The two Kinects are placed to preserve substantial overlap in their fields of view. This is a direct consequence of the finding of Bu and Lee [27] that iterative-closest-point refinement fails between near-opposing views for lack of point-cloud overlap; near-orthogonal placement retains both triangulation baseline and overlap.

3.3 Calibration

3.3.1 Intrinsics and extrinsics

Intrinsic parameters for every colour and depth stream are estimated from ChArUco board observations. Extrinsic calibration follows the workspace-anchor principle: rather than expressing cameras relative to one another, each camera is calibrated against a ChArUco board rigidly fixed to the workspace, defining a common world frame W. For camera k this yields

$$\boldsymbol{T}_{C_k}^{W} \in \mathrm{SE}(3), \tag{3.1}$$

the transform from the camera frame C_k to the workspace frame W. Anchoring to the workspace rather than to a reference camera has two consequences: adding or moving a camera does not invalidate the calibration of the others, and the robot base can be registered into the same frame without a separate camera-to-camera chain.

3.3.2 Robot registration

The manipulator base frame R is registered to W by observing a ChArUco target held in a known end-effector pose across a set of configurations, giving $\boldsymbol{T}_R^W$ directly. Where an explicit camera-to-robot transform is required instead, the Daniilidis dual-quaternion solver is used, following the robustness evidence of [33], with fifteen to twenty poses spanning a hemisphere.

3.4 Synchronisation

The two Azure Kinects are hardware-synchronised in master/subordinate configuration, with subordinate devices started before the master and depth exposures staggered to avoid mutual laser interference [30]. The RealSense cannot share that trigger; its synchronisation mode is incompatible at the electrical level.

Cross-family alignment therefore uses a single host-side monotonic clock. Every frame, from every device, is stamped on arrival with

$$\tau = \texttt{time.monotonic\_ns()}, \tag{3.2}$$

and streams are aligned by nearest neighbour in τ. Two points deserve emphasis. First, per-device hardware timestamps are *not* used for cross-device arithmetic: distinct device families count their own cycles from their own epochs, and the vendor documentation states explicitly that equal timestamps across cameras are not guaranteed [30]. Second, a monotonic clock rather than wall-clock time is required, since wall-clock time may be adjusted discontinuously by time synchronisation daemons, which would corrupt interval computations mid-recording.

Residual alignment jitter is measured rather than assumed (Section 4.4), and hardware synchronisation of the Kinect pair is verified by confirming bounded drift in the inter-camera timestamp difference [32].

3.5 Perception and Confidence-Weighted Fusion

3.5.1 Landmark detection and lifting

Hand landmarks are detected in each colour stream, yielding for view k at time t a set of image-plane landmarks with associated visibility scores $v_{i,t}^k \in [0, 1]$. Each landmark is lifted to three dimensions using the registered depth map rather than a learned depth estimate, exploiting the time-of-flight measurement directly, and transformed into the workspace frame by (3.1).

3.5.2 Fusion

Estimates from multiple views are combined by confidence-weighted averaging rather than by naive mean. For landmark i at time t, the fused position is

$$\hat{\boldsymbol{x}}_{i,t} \;=\; \frac{\sum_k w_{i,t}^k \, \boldsymbol{x}_{i,t}^k}{\sum_k w_{i,t}^k}, \tag{3.3}$$

with weights

$$w_{i,t}^k \;=\; \underbrace{v_{i,t}^k}_{\text{detector}} \cdot \underbrace{c_{i,t}^k}_{\text{sensor}} \cdot \underbrace{\frac{1}{\big(d_{i,t}^k\big)^2}}_{\text{range}} \cdot \underbrace{\max\big(0, \cos\theta_{i,t}^k\big)}_{\text{incidence}}, \tag{3.4}$$

where

$v_{i,t}^k$	detector visibility score for landmark i in view k;
$c_{i,t}^k$	sensor depth confidence or validity flag at that pixel;
$d_{i,t}^k$	range from camera k to the landmark;
$\theta_{i,t}^k$	angle between the local surface normal and the viewing direction.

The range and incidence factors follow established practice in geometric depth fusion, where surfaces observed near perpendicular are weighted above obliquely observed ones because they are measured more accurately [34]; the detector and sensor factors follow confidence-map re-weighting [35]. The product form is chosen so that a zero in any factor—an undetected landmark, an invalid depth pixel, a grazing view— removes that view's contribution entirely rather than merely attenuating it.

3.5.3 Wrist pose and gripper state

The wrist pose $\boldsymbol{T}_{H,t}^W \in \mathrm{SE}(3)$ is obtained by fitting an orthonormal frame to the fused wrist and knuckle landmarks. The gripper state is derived from the thumb–index distance,

$$g_t \;=\; \mathbb{1}\big[\, \|\hat{\boldsymbol{x}}_{\text{thumb},t} - \hat{\boldsymbol{x}}_{\text{index},t}\| < \tau_g \,\big]\,, \tag{3.5}$$

with hysteresis on τ_g to suppress chatter at the threshold. This single scalar is the entire hand representation beyond the wrist frame, following the convention established for parallel-jaw targets in [2, 4].

3.6 Object-Centric Representation

Let $\boldsymbol{T}^{W}_{O,t}$ denote the pose of the manipulated object in the workspace frame. Each demonstration is stored in two forms. The workspace-anchored form is $\boldsymbol{T}^{W}_{H,t}$ directly; the object-relative form is

$$\boldsymbol{T}^{O}_{H,t} \;=\; \left(\boldsymbol{T}^{W}_{O,t}\right)^{-1} \boldsymbol{T}^{W}_{H,t}. \tag{3.6}$$

The object-relative form is what transfers. If the same task is to be performed with the object at a new pose $\boldsymbol{T}^{R}_{O'}$ expressed in the robot frame, the desired end-effector pose at time t is

$$\boldsymbol{T}^{R,\mathrm{des}}_{E,t} \;=\; \boldsymbol{T}^{R}_{O'}\, \boldsymbol{T}^{O}_{H,t}\, \boldsymbol{T}^{H}_{E}\,, \tag{3.7}$$

where $\boldsymbol{T}^{H}_{E}$ is a fixed offset relating the human wrist frame to the robot end-effector frame, determined once per embodiment. Trajectories are retained densely rather than reduced to keyframes, since the continuous velocity profile of the demonstration carries task-relevant information— approach speed, contact timing—that sparse keyframes discard.

Both representations are exported. Storing both costs little and defers to the downstream consumer a choice that the literature does not settle universally: object-relative trajectories generalise across object placement [9, 12] but depend on object pose estimation, whereas workspace-anchored trajectories are simpler and adequate within a fixed layout.

3.7 Retargeting as a Single Cost Functional

3.7.1 Formulation

Let $\boldsymbol{q}_t \in \mathbb{R}^n$ be the manipulator configuration at time t and $f(\boldsymbol{q}_t) \in \mathrm{SE}(3)$ the forward kinematics of the end-effector. The task-space residual with respect to the desired pose (3.7) is the logarithmic map of the pose error,

$$\boldsymbol{e}_t(\boldsymbol{q}_t) \;=\; \mathrm{Log}\Big(\left(\boldsymbol{T}^{R,\mathrm{des}}_{E,t}\right)^{-1} f(\boldsymbol{q}_t)\Big)^{\vee} \in \mathbb{R}^6. \tag{3.8}$$

Retargeting is then posed as the single constrained problem

$$\begin{aligned}\min_{\boldsymbol{q}_{1:T}} \;\; J(\boldsymbol{q}_{1:T}) \;=\; \sum_{t=1}^{T} \Big[\; & \underbrace{\omega_t\, \rho_\delta(\|\boldsymbol{e}_t(\boldsymbol{q}_t)\|_{\boldsymbol{\Sigma}})}_{\text{data fidelity}} + \underbrace{\lambda_v \left\|\boldsymbol{q}_t - \boldsymbol{q}_{t-1}\right\|^2}_{\text{velocity}} \\ & + \underbrace{\lambda_a \left\|\boldsymbol{q}_t - 2\boldsymbol{q}_{t-1} + \boldsymbol{q}_{t-2}\right\|^2}_{\text{acceleration}} + \underbrace{\lambda_r \left\|\boldsymbol{q}_t - \boldsymbol{q}_{\mathrm{ref}}\right\|^2}_{\text{posture}} \; \Big]\end{aligned} \tag{3.9}$$

subject to

$$\boldsymbol{q}_{\text{min}} \;\le\; \boldsymbol{q}_t \;\le\; \boldsymbol{q}_{\text{max}}, \tag{3.10}$$

$$\|\boldsymbol{q}_t - \boldsymbol{q}_{t-1}\| \;\le\; \dot{\boldsymbol{q}}_{\text{max}}\,\Delta t, \tag{3.11}$$

$$h_j(\boldsymbol{q}_t) \;\ge\; 0, \quad j = 1, \dots, m, \tag{3.12}$$

where

ω_t	per-frame perception confidence weight, defined in (3.13);
ρ_δ	Huber loss with transition parameter δ;
$\boldsymbol{\Sigma}$	diagonal matrix normalising translational and rotational residual units;
$\lambda_v, \lambda_a, \lambda_r$	weights on velocity, acceleration and posture regularisation;
h_j	collision and self-collision barrier functions.

3.7.2 Confidence weighting

The per-frame weight aggregates the landmark confidences that produced the wrist estimate at time t,

$$\omega_t \;=\; \left(\frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \max_k w_{i,t}^k \right)^{\alpha}, \tag{3.13}$$

with $\mathcal{I}$ the landmark subset defining the wrist frame and α controlling how sharply low-confidence frames are discounted. The construction is the offline analogue of the distance-dependent switching weights used in online teleoperation retargeting [2]: frames in which the hand is partially occluded or poorly observed exert proportionally less influence on the solution, and the smoothness terms carry the trajectory through them instead of the data term dragging it toward a spurious observation.

3.7.3 Robustification

The Huber loss ρ_δ bounds the influence of gross outliers—a mis-detected hand, a depth dropout that survives the confidence weighting—at the level of the data term itself, rather than allowing a filtering stage to smear the outlier across neighbouring frames [6]. This is the mechanism that replaces pre-smoothing: outliers are down-weighted where they enter, not attenuated after they have already propagated.

3.7.4 Why a single functional

Equation (3.9) differs from the sequential alternative in one structurally important respect. The smoothness penalties act on $\boldsymbol{q}_t$, in the space where smoothness is required, and they are traded against data fidelity by the solver rather than applied blindly beforehand. In the sequential pipeline, task-space filtering cannot guarantee joint-space smoothness because the forward kinematics is nonlinear, and it cannot distinguish task-relevant fast motion from noise because it operates without reference to the data term. Here the trade-off is explicit: on high-confidence frames ω_t is large and the solution tracks the observation; on low-confidence frames the smoothness terms dominate and the solution interpolates.

Second-order smoothness, expressed by the acceleration term, follows the C^2-continuity argument of DexFlow [5]. A trajectory that is first-order smooth may still exhibit curvature discontinuities that appear as jerk on the physical manipulator and as label noise to a policy trained on the data.

3.7.5 Solution

The problem is solved with Pink on Pinocchio [8]. Each term of (3.9) maps onto a weighted task in the quadratic program—a frame task for the data term, posture and velocity regularisation tasks for the smoothness terms—while (3.10)–(3.12) enter as inequality constraints, with collision avoidance expressed through control barrier functions. Levenberg–Marquardt damping stabilises the solution near singularities.

Because the setting is offline, the problem is solved over the whole trajectory rather than frame by frame, so the acceleration term is well defined at every interior timestep and the solution is not constrained by the causality requirement that an online solver faces. The trajectory is initialised from the previous solution warm-started along t, and boundary conditions at $t = 1, 2$ are handled by replicating the initial configuration.

3.7.6 Interpolation

Where gaps remain after fusion—landmarks lost across all views—the trajectory is completed by cubic spline interpolation over $\mathrm{SE}(3)$ prior to optimisation. Dynamic filters that impose a motion model, such as Kalman or extended Kalman filters, are deliberately not applied: they introduce a prior about the dynamics that is subsequently re-estimated by the optimiser, and the interaction between the two priors is difficult to characterise. The optimiser already imposes the required regularity through λ_v and λ_a.

3.8 Replay Validation

Synthesised trajectories are executed on the physical manipulator before the dataset is finalised. The stage serves three purposes.

First, it filters infeasibility. A trajectory may satisfy (3.10)–(3.12) in the kinematic model and still fail on hardware through controller tracking limits, unmodelled collisions or workspace boundary conditions. Such episodes are rejected rather than exported.

Second, it closes the embodiment gap at the source. The proprioceptive states recorded during replay are those the robot actually attained, not those the optimiser predicted. Exporting the recorded rather than the synthesised states means the dataset contains honest observation–action pairs from the target embodiment, which is precisely the property that teleoperated datasets possess by construction and that naive video-derived datasets lack.

Third, it provides a curation signal. Episodes are ranked by tracking residual during replay, and the ranking is available to downstream consumers as a quality score, in the spirit of rollout-based demonstration curation [14] but applied before rather than after policy training.

Failure of an episode triggers either rejection or re-solution of (3.9) with modified weights—typically an increased λ_v or a relaxed data weight in the region of failure—as indicated by the feedback edge in Fig. 3.1. The number of re-solution attempts is bounded to prevent overfitting the trajectory to the replay controller.

3.9 Dataset Export

Surviving episodes are written in a LeRobot-compatible HDF5 layout. Each episode contains the synchronised camera streams, the recorded joint states and gripper states from replay, the object-relative and workspace-anchored wrist trajectories, per-frame confidence weights, and the replay tracking residual. Retaining the confidence weights and residuals in the exported artefact allows downstream curation to be repeated with different thresholds without re-recording.

3.10 Assumptions and Simplifications

The following assumptions delimit the scope of the work and are revisited in Section 5.4.

1. *Parallel-jaw target.* The hand is reduced to a wrist pose and a binary gripper state. Multi-fingered dexterous targets would require the finger-level representation rejected in Section 2.2.

2. *Rigid objects.* The object-relative representation (3.6) presupposes a rigid body with a well-defined pose. Deformable objects would require a flow-based representation [13].

3. *Single-arm, quasi-static tasks.* The task class is restricted to single-arm tabletop manipulation at moderate speed, where the quasi-static assumption underlying purely kinematic retargeting holds.

4. *Fixed workspace.* The ChArUco anchor is rigidly attached to the workspace and the camera rig is not moved within a recording session.

5. *Kinematic-only transfer.* Forces and contact dynamics are not estimated. Tasks whose success depends on force regulation are outside the scope.

Chapter 4

Implementation and Results

4.1 Software Architecture

The system is implemented in Python without a robotics middleware layer. Avoiding ROS was a deliberate choice: the pipeline is a batch data-processing system rather than a distributed real-time control system, and the node graph, message serialisation and build tooling that middleware provides would add operational complexity without addressing any requirement of the offline setting. Communication with the manipulator uses `ur-rtde` directly.

The implementation is organised into the layers listed in Table I. The separation follows the pipeline stages of Fig. 3.1, with the coordinate-frame contract—every spatial quantity carries an explicit frame identifier—enforced at module boundaries so that frame errors surface as type errors rather than as silent numerical mistakes.

The Azure Kinect depth engine requires a software rendering context inside the containerised deployment; this is obtained by forcing software rasterisation through the appropriate driver environment variables. The system is released under the MIT licence.

4.2 Experimental Setup

4.2.1 Hardware configuration

Experiments used a UR3 manipulator with a parallel-jaw gripper, two Azure Kinect DK cameras in master/subordinate hardware synchronisation and one Intel RealSense D435i, all observing a tabletop workspace anchored by a ChArUco board. Camera placement was near-orthogonal to preserve view overlap (Section 3.2). Recording ran at **XX** Hz.

TABLE I
Principal Modules of the Implementation

Module	Responsibility	Principal dependencies
`capture`	Device drivers, hardware sync, host-clock stamping	Azure Kinect SDK, `pyrealsense2`
`calibration`	ChArUco detection, intrinsics, extrinsics, robot registration	OpenCV
`perception`	Landmark detection, depth lifting, confidence-weighted fusion	MediaPipe, OpenCV
`representation`	Object pose, object-relative transform, spline interpolation	NumPy, SciPy
`retargeting`	Cost functional assembly and solution	Pink, Pinocchio
`replay`	Trajectory execution, proprioception logging, screening	`ur-rtde`
`export`	LeRobot/HDF5 serialisation	h5py, LeRobot
`interface`	Recording control and inspection	FastAPI, Streamlit

4.2.2 Task suite

Evaluation used a deliberately constrained task class, chosen so that the quasi-static and rigid-object assumptions of Section 3.10 hold and so that success is unambiguously measurable:

1. *Pick-and-place.* Grasp a rigid object and place it at a marked target location.
2. *Pick-and-place with displacement.* As above, but with the object initialised at a pose not seen during demonstration, to test the object-relative representation.
3. *Pouring.* Grasp a container and tip it over a target, testing orientation fidelity rather than position alone.
4. **[Additional task, if included.]**

For each task, **N** demonstrations were recorded by **one** demonstrator.

4.2.3 Baselines and ablations

The proposed formulation (3.9) is compared against:

1. **B1 — Sequential.** Task-space low-pass filtering, per-frame inverse kinematics, joint-space low-pass filtering. This is the pattern used by offline pipelines such as [10] and the principal comparison for H1.

2. **B2 — Joint, no confidence.** Equation (3.9) with $\omega_t \equiv 1$. Isolates the contribution of confidence weighting for H2.

3. **B3 — Joint, no acceleration.** Equation (3.9) with $\lambda_a = 0$. Isolates the contribution of second-order smoothness.

4. **B4 — Joint, no Huber.** Squared loss in place of ρ_δ. Isolates outlier robustification.

5. **B5 — World-frame.** Workspace-anchored rather than object-relative targets, evaluated on the displaced-object task only.

4.3 Evaluation Protocol and Metrics

Metrics were fixed before the final experimental runs, so that the criteria of success were not selected after observing the outcomes.

4.3.1 Retargeting fidelity

Task-space tracking error between the desired end-effector pose (3.7) and the achieved pose $f(\boldsymbol{q}_t)$, reported separately for translation and rotation:

$$\mathrm{RMSE}_{\mathrm{pos}} = \sqrt{\frac{1}{T}\sum_t \left\|\boldsymbol{p}_t^{\mathrm{des}} - \boldsymbol{p}_t\right\|^2}, \qquad \mathrm{RMSE}_{\mathrm{rot}} = \sqrt{\frac{1}{T}\sum_t \left\|\mathrm{Log}(\boldsymbol{R}_t^{\mathrm{des}\,\top}\boldsymbol{R}_t)\right\|^2}. \tag{4.1}$$

4.3.2 Smoothness

Spectral arc length of the joint velocity profile, a dimensionless smoothness measure that is invariant to amplitude and duration, together with mean absolute joint jerk. Lower magnitudes indicate smoother motion.

4.3.3 Feasibility

The proportion of synthesised episodes that execute on hardware without controller fault, joint limit violation or collision, and the mean tracking residual reported by the controller during replay.

4.3.4 Downstream policy success

ACT and Diffusion Policy trained on the exported dataset and evaluated over **N** rollouts per task, reporting success rate under a task-specific success predicate.

4.3.5 Calibration and sensing accuracy

Reprojection error of the ChArUco calibration in pixels; cross-camera registration error as the mean distance between corresponding points in overlapping regions; and residual timestamp misalignment between device families.

4.4 Calibration and Sensing Results

TABLE II
Calibration and Synchronisation Accuracy

Quantity	Measured	Reference from literature
ChArUco reprojection error (px)	**X.XX**	< 0.5 typical
Cross-camera registration, IR (mm)	**XX.X**	7.4–9.9 [24]
Cross-camera registration, colour (mm)	**XX.X**	20.9–21.4 [24]
Kinect pair timestamp drift (ms/min)	**X.XX**	< 1 [32]
Kinect–RealSense residual offset (ms)	**X.XX**	—
Wrist position repeatability (mm)	**X.XX**	—

[TODO: Populate Table II from the calibration runs; discuss whether measured cross-camera registration falls within the range reported by [24], and if not, whether the discrepancy is attributable to board size, view overlap or working distance.]

4.5 Retargeting Fidelity

TABLE III
Retargeting Fidelity and Smoothness, Proposed Formulation Versus Sequential Baseline

Method	RMSE_{pos} (mm)	RMSE_{rot} (deg)	SAL	Jerk (rad/s^3)
B1 Sequential	**XX.X**	**X.X**	**−X.XX**	**XX.X**
Proposed (3.9)	**XX.X**	**X.X**	**−X.XX**	**XX.X**
Relative change	**−XX%**	**−XX%**	**−XX%**	**−XX%**

[placeholder — insert figure]

Fig. 4.1. Joint trajectory for a representative pick-and-place episode under the sequential baseline (B1) and the proposed formulation. [TODO: insert plot]

[TODO: Report whether H1 is supported. State the statistical test used and the number of episodes; if the improvement is present on one metric but not the other, say so plainly rather than reporting only the favourable one.]

4.6 Ablation Studies

TABLE IV
Ablation of the Individual Terms of the Cost Functional

Variant	Removed term	$\mathrm{RMSE}_{\mathrm{pos}}$ (mm)	SAL
Proposed	—	**XX.X**	**−X.XX**
B2	confidence weighting ω_t	**XX.X**	**−X.XX**
B3	acceleration penalty λ_a	**XX.X**	**−X.XX**
B4	Huber robustification ρ_δ	**XX.X**	**−X.XX**

TABLE V
Effect of Confidence Weighting Stratified by Landmark Occlusion

Frame subset	B2 (uniform)	Proposed (weighted)
All frames	**XX.X**	**XX.X**
Low-occlusion frames	**XX.X**	**XX.X**
High-occlusion frames	**XX.X**	**XX.X**

Values are $\mathrm{RMSE}_{\mathrm{pos}}$ in mm. Occlusion stratification uses the median of ω_t as the threshold.

[TODO: Report whether H2 is supported, and specifically whether the benefit of confidence weighting is larger on high-occlusion frames as hypothesised. If the effect is uniform across strata, the mechanism proposed in Section 3.7 is not the operative one and this must be stated.]

4.7 Replay Feasibility

[TODO: Report the distribution of rejection causes—joint limit, singularity proximity, collision, controller tracking fault—since the distribution indicates which constraint in (3.10)–(3.12) is insufficiently modelled rather than merely how many episodes failed.]

TABLE VI
Outcome of Physical Replay Screening

Task	Episodes synthesised	Rejected at replay	Rejection cause (dominant)
Pick-and-place	**NN**	**NN** (**XX%**)	**[cause]**
Pick-and-place (displaced)	**NN**	**NN** (**XX%**)	**[cause]**
Pouring	**NN**	**NN** (**XX%**)	**[cause]**
Total	**NN**	**NN** (**XX%**)	—

TABLE VII
Success Rate of Policies Trained on the Exported Dataset

Task	ACT	Diffusion Policy
Pick-and-place	**XX%**	**XX%**
Pick-and-place (displaced)	**XX%**	**XX%**
Pouring	**XX%**	**XX%**
Mean	**XX%**	**XX%**

TABLE VIII
Effect of Representation Choice on the Displaced-Object Task

Representation	Success rate
B5 Workspace-anchored (world frame)	**XX%**
Object-relative (3.6)	**XX%**

4.8 Downstream Policy Performance

[TODO: Report whether H3 is supported by comparing policy success on datasets with and without replay screening. Note that this requires training on the unscreened dataset as well, which doubles the training budget; if that comparison is not run, H3 must be evaluated on the executability metric alone and the weaker claim stated explicitly.]

Chapter 5

Analysis and Discussion

5.1 Interpretation of the Results

[TODO: This section interprets rather than restates Chapter 4. Write it after the numbers are in, structured as: what the key findings mean for each hypothesis; which results support them and which do not; how the findings compare with prior work; explanation of discrepancies; unexpected findings.]

The three hypotheses of Section 1.4 address separable mechanisms, and the argument for each is independent of the others.

H1 — formulation. The claim is that posing retargeting as one weighted functional dominates the sequential pipeline on both fidelity and smoothness simultaneously. The distinction matters because the sequential pipeline can be made arbitrarily smooth by increasing filter strength, at the cost of fidelity; the interesting question is therefore not whether either metric can be improved in isolation but whether the trade-off frontier itself moves. **[Insert result.]** If the proposed method improves one metric while degrading the other, the correct conclusion is that the frontier has not moved and the improvement is a re-parameterisation of the same trade-off.

H2 — confidence weighting. The mechanism proposed in Section 3.7 predicts a specific signature: the benefit of weighting should concentrate on frames where perception is degraded, because that is where uniform weighting drags the solution toward a bad observation. A uniform improvement across occlusion strata would indicate that the weighting is acting as a global regulariser rather than through the proposed mechanism. **[Insert result.]**

H3 — replay screening. The claim is about dependability rather than accuracy: screening should raise the executable fraction of the exported dataset. **[Insert result.]** The stronger claim—that screening improves downstream policy success—requires the paired training run noted in Section 4.8 and should not be asserted on the executability metric alone.

5.2 Comparison with Prior Work

Direct numerical comparison with the systems reviewed in Chapter 2 is not meaningful, since none is evaluated on the same task suite, embodiment or sensing configuration. What can be compared is qualitative behaviour and the class of guarantees provided.

Relative to OKAMI [10], the present system discards the parametric body-and-hand reconstruction and the two-stage warp-then-solve structure. The expected consequence is a lower perception error floor—time-of-flight depth in place of a mesh fit—at the cost of being restricted to parallel-jaw targets. Relative to ORION [9], the present system retains the dense wrist trajectory that ORION discards, which should preserve velocity-profile information at the cost of a dependency on reliable hand observation. Relative to the online teleoperation systems [1, 2, 4], the formulation is the same but the operating regime is not: the offline setting permits whole-trajectory optimisation with a well-defined acceleration term at every interior timestep and admits a physical validation stage, neither of which an interactive-rate solver can perform.

The measured cross-camera registration error (Table II) should be compared against the 7.4–21.4 mm range reported by Romeo et al. [24]. [TODO: State whether the measured value falls in that range and, if not, diagnose the cause rather than reporting the discrepancy without explanation.]

5.3 Error Budget

A central methodological commitment of this work is that residual error is decomposed rather than reported as a single aggregate figure. Three contributions are independent in origin and demand different remedies; conflating them would make the aggregate uninformative about what to improve.

1. *Landmark detection noise.* Error in the image-plane hand landmarks, propagated through depth lifting. This contribution is reduced by confidence weighting (3.4) and by multi-view fusion, and it is characterised by the wrist-position repeatability measurement of Table II. It is not reduced by better calibration.

2. *Calibration error.* Error in the camera-to-workspace and robot-to-workspace transforms (3.1). This contribution is systematic within a session, appears as a constant offset rather than as noise, and is characterised by the reprojection and cross-camera registration measurements. It is not reduced by multi-view fusion, since all views inherit it.

3. *Representation dependency.* Error introduced by the choice of reference frame, specifically the sensitivity of world-frame trajectories to object displacement. This contribution is zero when the object is where it was during demonstration and grows with displacement; it is characterised by the comparison of Table VIII rather than by any calibration measurement.

TABLE I
Decomposition of Residual Wrist-Pose Error by Source

Source	Character	Magnitude (mm)	Reduced by
Landmark detection	random	**X.X**	fusion, confidence weighting
Calibration	systematic	**X.X**	board size, view count
Representation	conditional	**X.X**	object-relative frame
Combined	—	**XX.X**	—

[TODO: Populate and discuss. If the combined figure differs materially from the quadrature sum of the independent contributions, the sources are not independent and the decomposition must be revised rather than presented as if it held.]

5.4 Limitations

The assumptions of Section 3.10 translate into the following limitations, which are stated plainly rather than deferred to future work.

1. *Gripper abstraction.* A binary gripper state cannot represent in-hand manipulation, regrasping or force-modulated grasps. Tasks requiring these are outside the demonstrated capability, and extending to dexterous hands would require reinstating the finger-level representation that Section 2.2 rejects.

2. *Rigid-object assumption.* The object-relative transform (3.6) is undefined for deformable objects. Cloth, cable and granular manipulation are excluded.

3. *Dependence on object pose estimation.* The object-relative representation inherits the error of the object pose estimator. Where that estimator fails—textureless, symmetric or occluded objects—the representation degrades to worse than the world-frame alternative it is meant to improve upon.

4. *Kinematic transfer only.* No force or contact model is estimated. Success on tasks with a force-regulation component is not claimed.

5. *Single demonstrator, constrained task class.* The evaluation uses **one** demonstrator and **three** task types. Generalisation across demonstrators, hand morphologies and task families is not established.

6. *Sensing floor.* Cross-camera agreement at the centimetre level [24] bounds achievable accuracy. Tasks with tolerances below that bound cannot be addressed by this sensing configuration regardless of the retargeting formulation.

7. *Cross-family synchronisation.* Alignment between the Kinect pair and the RealSense is software-based and therefore millisecond-scale; fast motion may exhibit residual misalignment between those streams.

5.5 Threats to Validity

Internal validity. Metrics were fixed before the final runs (Section 4.3) to prevent selection of criteria after observing outcomes. Baselines B1–B5 differ from the proposed method in exactly one component each, so attribution of any observed difference is unambiguous. The principal residual threat is that the sequential baseline's filter parameters are themselves tunable; an unfavourable choice would make the comparison misleading. [TODO: State how B1 filter parameters were selected—ideally by tuning them to optimise the baseline's own fidelity, not by adopting a default.]

External validity. The constrained task class was chosen for measurability, which necessarily limits generalisation. Results should not be read as claims about manipulation in general.

Construct validity. Spectral arc length measures smoothness of the velocity profile, which is a proxy for the property actually of interest—that a policy can imitate the trajectory. The proxy is standard but imperfect, and the downstream policy results of Section 4.8 are the more direct evidence.

5.6 Implications

Three implications extend beyond the specific system.

First, the transferability of the online retargeting formulation to the offline regime suggests that the separation between the teleoperation and video-imitation literatures is a historical accident rather than a technical necessity. The offline setting is strictly less constrained—no causality requirement, no latency budget—and therefore able to use the same formulation more thoroughly than the setting for which it was designed.

Second, moving curation upstream of export, so that infeasible trajectories are removed before they enter the dataset rather than filtered after they have degraded a policy, is applicable to any pipeline that synthesises rather than records actions. The cost is one physical execution per episode; the benefit is that the exported dataset contains proprioception the robot actually attained.

Third, the error decomposition of Section 5.3 is a transferable methodological point. Reporting a single aggregate accuracy figure for a multi-stage perception pipeline obscures which stage limits performance, and consequently which engineering effort would pay.

Chapter 6

Conclusion

6.1 Summary

This thesis has addressed the question of how manipulation demonstrations recorded passively, with a human using their own hands and no teleoperation interface, can be converted into datasets suitable for training visuomotor imitation policies. The work has argued that the two established approaches to this problem are each incomplete: online teleoperation systems have converged on a sound retargeting formulation—a single weighted cost functional combining data fidelity, temporal smoothness and kinematic constraints—but produce no dataset, while offline video-to-robot pipelines produce datasets but retarget either not at all, through sparse object keyframes, or through a sequential filter–solve–filter pattern whose smoothing is applied in the wrong space to guarantee the property it is intended to provide.

ViKi has been presented as a system occupying the intersection. It captures demonstrations with a metrically calibrated multi-camera RGB-D rig anchored to a ChArUco workspace frame; represents the hand minimally, as a wrist pose in $\mathrm{SE}(3)$ with a binary gripper state; stores each demonstration both relative to the manipulated object and relative to the workspace anchor; retargets by minimising one weighted functional in which the data term is scaled by per-frame perception confidence and robustified by a Huber loss, and in which velocity and acceleration penalties act directly in joint space; validates the result by executing it on the physical manipulator before export; and writes a LeRobot-compatible dataset containing the proprioception the robot actually attained rather than the states the optimiser predicted.

6.2 Answers to the Research Questions

RQ1. **[State the measured effect of the single-functional formulation on fidelity and smoothness relative to the sequential baseline, and whether the trade-off frontier moved rather than merely shifted along.]**

RQ2. **[State the achieved wrist-pose accuracy and how the residual decomposes across landmark noise, calibration error and representation dependency, per Table I.]**

RQ3. **[State the proportion of episodes rejected by replay screening, the dominant rejection causes, and the resulting change in the executable fraction of the exported dataset.]**

6.3 Contributions

The contributions of this work are the transfer of the single-cost-functional retargeting formulation from the online teleoperation regime, for which it was developed, into the offline object-centric dataset-generation regime, where the review of Chapter 2 found it unused; the coupling of that formulation to a metrically calibrated multi-view RGB-D front end with confidence-weighted fusion and explicit cross-device timestamp handling; the use of physical replay as a pre-export dataset-curation stage rather than a post-training evaluation; and an error decomposition that separates the three independent contributions to residual pose error rather than reporting an aggregate.

It is worth restating what is *not* claimed. The cost functional is not novel as an operator; its components appear in [1, 5, 2, 6], and a claim of algorithmic novelty would be refuted by that prior art. Object-centric representation, ChArUco anchoring, confidence-weighted depth fusion and host-clock synchronisation are all established practice. The contribution is the conjunction and the regime transfer, which the positioning analysis of Section 2.7 found unoccupied.

6.4 Limitations

The system is restricted to parallel-jaw targets, rigid objects, single-arm quasi-static tabletop tasks and a fixed workspace; it performs kinematic transfer only, estimating no forces; its accuracy is bounded below by centimetre-level cross-camera agreement; and its evaluation uses a single demonstrator on a constrained task class. These are stated in full in Section 5.4.

6.5 Future Work

Several directions follow from the limitations.

1. *Dexterous targets.* Extending beyond a binary gripper requires reinstating a finger-level representation. The ablation evidence of [7] suggests that the wrist objective should remain weakly weighted relative to fingertip objectives, which would change the balance of terms in (3.9) rather than its structure.

2. *Deformable objects.* Replacing the rigid object pose of (3.6) with a flow-based object representation [13] would remove the rigidity assumption at the cost of a less compact transfer rule.

3. *Closing the replay loop.* Currently, replay failures trigger bounded re-solution with hand-adjusted weights. Learning the weight adjustment from the failure mode would make the loop automatic.

4. *Cross-embodiment transfer.* The object-relative representation is embodiment-agnostic by construction; the same recorded demonstrations could be retargeted to the KUKA iiwa14 available in the laboratory, and the degree to which policy performance transfers would test the representational claim directly.

5. *Scaling demonstrator diversity.* Recording from multiple demonstrators with differing hand morphologies would test whether the fixed hand-to-end-effector offset $\boldsymbol{T}_E^H$ of (3.7) requires per-demonstrator calibration.

6. *Downstream curation.* The confidence weights and replay residuals retained in the exported dataset support curation experiments in the manner of [14] without re-recording; whether the replay residual is a better curation signal than a learned rollout classifier is an open question.

6.6 Concluding Remarks

The practical claim of this work is modest and, for that reason, defensible: a formulation already accepted in one part of the robot-learning literature works in another part where it had not been applied, and the infrastructure required to apply it—metric multi-view sensing, honest cross-device synchronisation, physical validation before export—is buildable at the scale of a single laboratory. If the results hold, the cost of producing imitation-learning data falls without the cost being transferred to policy robustness, which is the trade that passive capture has historically demanded.

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Note: entries marked in this colour require completion or verification before submission. Preprint entries dated 2026 could not be confirmed as peer-reviewed at the time of writing and are cited as preprints.

Appendix A

Notation

TABLE I
Symbols Used Throughout the Thesis

Symbol	Meaning
W	workspace frame, defined by the ChArUco anchor
C_k	frame of camera k
R	manipulator base frame
E	manipulator end-effector frame
H	human wrist frame
O	manipulated object frame
$\boldsymbol{T}_B^A$	rigid transform from frame B to frame A, element of SE(3)
$\boldsymbol{q}_t \in \mathbb{R}^n$	manipulator configuration at timestep t
$f(\boldsymbol{q}_t)$	forward kinematics of the end-effector
$\boldsymbol{e}_t$	task-space residual, (3.8)
ω_t	per-frame perception confidence weight, (3.13)
$w_{i,t}^k$	per-landmark fusion weight, (3.4)
$v_{i,t}^k$	detector visibility score
$c_{i,t}^k$	sensor depth confidence
ρ_δ	Huber loss with transition parameter δ
$\lambda_v, \lambda_a, \lambda_r$	velocity, acceleration and posture weights
g_t	binary gripper state, (3.5)
τ	host-side monotonic timestamp, (3.2)
T	number of timesteps in an episode

Appendix B

Exported Dataset Schema

Each episode is stored as an HDF5 group with the fields listed in Table I. The layout is compatible with LeRobot conventions; fields beyond the LeRobot minimum are retained so that curation can be repeated downstream without re-recording.

TABLE I
Per-Episode Dataset Fields

Field	Shape	Description
`observation.images.*`	$T \times H \times W \times 3$	per-camera colour streams
`observation.state`	$T \times n$	recorded joint positions from replay
`action`	$T \times (n{+}1)$	commanded joint positions and gripper
`wrist_pose_world`	$T \times 4 \times 4$	$\boldsymbol{T}^{W}_{H,t}$
`wrist_pose_object`	$T \times 4 \times 4$	$\boldsymbol{T}^{O}_{H,t}$, (3.6)
`object_pose_world`	$T \times 4 \times 4$	$\boldsymbol{T}^{W}_{O,t}$
`gripper_state`	T	binary, (3.5)
`confidence`	T	ω_t, (3.13)
`replay_residual`	T	controller tracking residual during replay
`timestamp_host`	T	host monotonic timestamp, (3.2)

Appendix C

Solver Configuration

TABLE I
Weights and Solver Parameters Used in the Reported Experiments

Parameter	Value	Selection method
λ_v (velocity)	**X.XX**	**[grid search / held-out set]**
λ_a (acceleration)	**X.XX**	**[...]**
λ_r (posture)	**X.XX**	**[...]**
α (confidence sharpness)	**X.XX**	**[...]**
δ (Huber transition)	**X.XX**	**[...]**
τ_g (gripper threshold, mm)	**XX**	**[...]**
LM damping	**X.XX**	**[...]**
Recording rate (Hz)	**XX**	—
Max replay re-solutions	**X**	—

[TODO: Weight selection must be described honestly: if weights were tuned on the same episodes used for evaluation, say so and report the consequence, or re-tune on a held-out split.]